



The Fifth Unregulated Contaminant Monitoring Rule (UCMR 5) Data Summary: July 2026

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Overview

The U.S. Environmental Protection Agency released final set of data collected under the fifth Unregulated Contaminant Monitoring Rule ([UCMR 5](#)) for the 30 chemical contaminants (29 per- and polyfluoroalkyl substances [PFAS] and lithium) listed in [Table 1](#). The Safe Drinking Water Act (SDWA) requires that once every five years the EPA issue a list of unregulated contaminants to be monitored by public water systems (PWSs). The data collected under UCMR 5 improve understanding of the prevalence and amount of 29 PFAS and lithium in the nation's drinking water systems. To access the results, visit the [UCMR Occurrence Data](#) website.

On April 10, 2024, the EPA announced the final [National Primary Drinking Water Regulation \(NPDWR\)](#) for certain PFAS that are among the 29 PFAS monitored in UCMR 5, and established legally enforceable standards, or Maximum Contaminant Levels (MCLs), for these PFAS in drinking water. On May 18, 2026, the EPA announced two proposed rules related to PFAS in drinking water. The first rule proposes to uphold the NPDWR for PFOA and PFOS, while providing an option for PWSs to [request two additional years](#) to comply with the MCLs. The second rule proposes to [rescind the regulations](#) for PFHxS, PFNA, HFPO-DA (commonly known as GenX), and the Hazard Index mixture of these three PFAS plus PFBS. Visit the EPA's PFAS [Final NPDWR](#) and [NPDWR Implementation](#) websites for more information. **Information relevant to the April 2024 NPDWR provided in this document is for technical assistance only and does not supersede the NPDWR requirements in 40 CFR 141 and 40 CFR 142.**

UCMR 5 results for the NPDWR PFAS do not indicate compliance or noncompliance with the MCLs. PWSs may work with their state to submit their UCMR 5 monitoring data to meet some or all of the PFAS NPDWR's initial monitoring requirements. Compliance with the PFAS MCLs will be determined by calculating the running annual average (RAA) of quarterly results for each sample point based on compliance monitoring data reported to the primacy agency. Recognizing that the UCMR 5 results have no direct implications for PWS MCL compliance, but that many are interested in comparing UCMR 5 data to the MCLs, the EPA calculated RAAs for the UCMR 5 results where such a calculation was possible. The agency then compared those averages to the MCLs. The UCMR 5 dataset represents approximately 10,130 PWSs with full sets of results for approximately 25,950 sampling locations (*i.e.*, two sample results for locations with ground water sources or four sample results for locations with surface water sources).

The EPA notes the following for the complete dataset:

- The approximate percentages of PWSs that reported a full set of results for at least one location and had an average for one or more of the NPDWR PFAS that was greater than the respective MCL(s) are:
 - 7.1% of small PWSs serving fewer than 3,300 people,
 - 8.3% of small PWSs serving 3,300 to 10,000 people (hereafter referred to as medium PWSs¹), and
 - 15.5% of large PWSs serving more than 10,000 people.
- A weighted estimate of the total PWSs nationwide with one or more RAA greater than a PFAS NPDWR MCL is **7.8%**. This weighted estimate is provided since the UCMR 5 dataset includes only a subset of the PWSs serving fewer than 3,300 people that are subject to the requirements of the PFAS NPDWR. Therefore, a weighted estimate ensures the PWS size categories are appropriately represented relative to the amount of PWSs nationally and PWSs serving 3,300 or more people are not overrepresented.
- UCMR 5 data show that PFAS co-occur as mixtures in drinking water systems. 66% of sampling locations with at least one PFAS result at or above the UCMR 5 MRL had results for multiple PFAS at or above the

¹ Note: The Safe Drinking Water Information System Federal Reporting Services (SDWIS/Fed) database and PFAS NPDWR Economic Analysis define medium PWSs as those serving 3,301 to 10,000 people.

UCMR 5 MRL.

- The EPA established a non-regulatory health-based reference concentration for lithium (*i.e.*, EPA's health reference level [HRL]) for screening purposes. 28.5% of small PWSs serving 10,000 or fewer people and 24.1% of large PWSs serving more than 10,000 people reported lithium results above the reference concentration.

Regulatory levels (*e.g.*, MCLs), health-based reference values, and other contaminant health effects information are provided in [Table 2](#). Summary details for contaminant occurrence, including comparisons to the PFAS NPDWR MCLs and PFAS co-occurrence, are shown in [Table 3](#), [Table 4](#), and [Table 5](#). **This document represents the final UCMR 5 dataset. The information in this document is current as of July 2026 and will no longer be updated by the agency.**

Background

The EPA uses the UCMR program to collect nationally representative data for contaminants that may be present in drinking water but are not currently subject to regulatory standards set under SDWA. This monitoring is used by the agency to understand the frequency and level of occurrence of unregulated contaminants in the nation's PWSs. Every five years, taking into consideration the EPA's Contaminant Candidate List (CCL), the agency develops a new UCMR contaminants list for monitoring. SDWA, as amended by Section 2021 of America's Water Infrastructure Act of 2018, calls for the EPA to:

- Issue a list of unregulated contaminants to be monitored by certain PWS types¹ every five years.
- Require:
 - All large PWSs¹ serving more than 10,000 people to monitor their water for the contaminants.
 - All small PWSs¹ serving between 3,300 and 10,000 people to monitor, subject to the availability of EPA appropriations and sufficient laboratory capacity.
 - A nationally representative sample of small PWSs¹ serving fewer than 3,300 people to monitor.
- Pay for the analysis of UCMR samples from participating small PWSs serving 10,000 or fewer people.
- Make analytical results available in a National Contaminant Occurrence Database ([NCOD](#)) for drinking water.

UCMR 5 data collection helps inform the EPA's PFAS OUT initiative to enhance engagement with PWSs and share resources, tools, funding, and free [Real Water Technical Assistance \(RealWaterTA\)](#) to help PWSs address PFAS in drinking water. The EPA's RealWaterTA initiatives work with PWSs nationwide to identify affordable solutions to assess and address PFAS, including PFOA and PFOS. Services offered to PWSs include water quality testing, development of technical plans, operator training support, designing public engagement and outreach strategies, and support for accessing federal funding opportunities.

¹ UCMR 5 requirements applied to community water systems (CWSs) and non-transient non-community water systems (NTNCWSs). They did not apply to transient non-community water systems (TNCWSs). The use of "PWS" throughout this document refers to CWSs and NTNCWSs that participated in UCMR 5 monitoring. For more information on PWS types, visit the agency's [website](#).

Table 1. Contaminants and Methods

Contaminant ¹	CASRN ²	EPA Method	Contaminant Classification
lithium	7439-93-2	200.7	Metal/Pharmaceutical
perfluorooctanesulfonic acid (PFOS)	1763-23-1	533	PFAS
perfluorooctanoic acid (PFOA)	335-67-1	533	PFAS
hexafluoropropylene oxide dimer acid (HFPO-DA) (GenX chemicals)	13252-13-6	533	PFAS
perfluorohexanesulfonic acid (PFHxS)	355-46-4	533	PFAS
perfluorononanoic acid (PFNA)	375-95-1	533	PFAS
perfluorobutanesulfonic acid (PFBS)	375-73-5	533	PFAS
perfluorobutanoic acid (PFBA)	375-22-4	533	PFAS
perfluorohexanoic acid (PFHxA)	307-24-4	533	PFAS
perfluorodecanoic acid (PFDA)	335-76-2	533	PFAS
1H, 1H, 2H, 2H-perfluorooctane sulfonic acid (6:2 FTS)	27619-97-2	533	PFAS
1H, 1H, 2H, 2H-perfluorodecane sulfonic acid (8:2 FTS)	39108-34-4	533	PFAS
1H, 1H, 2H, 2H-perfluorohexane sulfonic acid (4:2 FTS)	757124-72-4	533	PFAS
4,8-dioxa-3H-perfluorononanoic acid (ADONA)	919005-14-4	533	PFAS
11-chloroeicosafluoro-3-oxaundecane-1-sulfonic acid (11Cl-PF3OUdS)	763051-92-9	533	PFAS
9-chlorohexadecafluoro-3-oxanone-1-sulfonic acid (9Cl-PF3ONS)	756426-58-1	533	PFAS
nonafluoro-3,6-dioxaheptanoic acid (NFDHA)	151772-58-6	533	PFAS
perfluoro (2-ethoxyethane) sulfonic acid (PFEEESA)	113507-82-7	533	PFAS
perfluoro-3-methoxypropanoic acid (PFMPA)	377-73-1	533	PFAS
perfluoro-4-methoxybutanoic acid (PFMBA)	863090-89-5	533	PFAS
perfluorododecanoic acid (PFDoA)	307-55-1	533	PFAS
perfluoroheptanesulfonic acid (PFHpS)	375-92-8	533	PFAS
perfluoroheptanoic acid (PFHpA)	375-85-9	533	PFAS
perfluoropentanesulfonic acid (PFPeS)	2706-91-4	533	PFAS
perfluoropentanoic acid (PFPeA)	2706-90-3	533	PFAS
perfluoroundecanoic acid (PFUnA)	2058-94-8	533	PFAS
n-ethyl perfluorooctanesulfonamidoacetic acid (NEtFOSAA)	2991-50-6	537.1	PFAS
n-methyl perfluorooctanesulfonamidoacetic acid (NMeFOSAA)	2355-31-9	537.1	PFAS
perfluorotetradecanoic acid (PFTA)	376-06-7	537.1	PFAS
perfluorotridecanoic acid (PFTrDA)	72629-94-8	537.1	PFAS

¹ UCMR 5 contaminants were monitored under the UCMR Assessment Monitoring (AM) design. For more information, visit the [UCMR 5 website](#).

² CASRN – Chemical Abstracts Service Registry Number

Information About UCMR 5 Results

The purpose of this document is to summarize UCMR 5 results and provide context around UCMR 5 results in relation to EPA-established UCMR minimum reporting levels (MRLs) and, if available, health-based reference values (*e.g.*, non-regulatory drinking water reference concentrations, reference doses [RfDs], cancer slope factors [CSFs]) and regulatory values (*e.g.*, MCLs).

The UCMR MRLs are the lowest concentrations that laboratories may report to the EPA for UCMR monitoring. UCMR MRLs are determined using data from multiple laboratories that participate in the EPA's MRL-setting studies and are not associated with contaminant health effects information. The EPA establishes UCMR MRLs to ensure consistency in the quality of the information reported to the agency.

Depending on the availability of a final human health toxicity assessment for a UCMR contaminant, a non-regulatory [health-based reference concentration](#) in drinking water may be available. Drinking water reference concentrations can be derived from an RfD (*i.e.*, a non-cancer endpoint) or an oral CSF (*i.e.*, a cancer endpoint), if available, and consider additional assumptions about body weight and drinking water intake. For non-cancer endpoints, the relative source contribution of drinking water to total intake is applied to derive the health-based drinking water reference concentration. Reference concentrations presented in this document do not represent regulatory limits or action levels, and should not be interpreted as an indication of future agency actions.

Community water systems (CWSs) required to monitor under UCMR must inform their customers of UCMR results (including the average and range of results) in their Consumer Confidence Report (CCR). See 40 CFR 141.153(d)(7) for the CCR regulatory requirements and the EPA's [CCR Compliance Help webpage](#) for resources. Non-transient non-community water systems (NTNCWSs) (*e.g.*, a school that operates its own drinking water system) and CWSs required to monitor under UCMR must inform their customers of the availability of UCMR results through Tier 3 Public Notification (PN). See 40 CFR 141.207 for the PN regulatory requirements and the EPA's [PN Compliance Help webpage](#) for resources. CCR and PN requirements apply to all UCMR results (including results for the UCMR 5 PFAS that are subject to the NPDWR) and are separate from the PFAS NPDWR CCR and PN requirements.

UCMR occurrence data are used to inform the agency's [Regulatory Determination](#) process. States may also establish requirements or levels (regulatory or non-regulatory) for drinking water contaminants. PWSs are responsible for being aware of and complying with their state's requirements, if any.

Health-Based Reference Values

[Table 2](#) provides health-based reference values (*i.e.*, non-regulatory health-based drinking water reference concentrations and RfDs) for each contaminant monitored under UCMR 5, if available, as of July 2026. To identify reference values, the EPA applied the following principles:

- (1) Reference concentrations and RfDs were compiled from the following publicly available resources:
 - a. EPA [Technical Support Document for the Final CCL 5 – Contaminant Information Sheets](#) and
 - b. EPA [Integrated Risk Information System \(IRIS\) Assessments](#)
 - c. EPA [Human Health Toxicity Value Assessments](#)
 - d. U.S. Geological Survey (USGS) [Health-Based Screening Levels \(HBSLs\)](#)

The above resources are the products (or compilation) of peer-reviewed health assessments. The reference values are subject to change as new health assessments are completed; they are not legally enforceable federal standards.

- (2) If health information was available from more than one of the resources listed above, the most recent health information was used.
- (3) If both cancer and non-cancer drinking water reference concentrations were available from the most recent resource, the lower of the two concentrations was used. Review the references and footnotes in [Table 2](#) for additional health effects information.
- (4) If an RfD (*i.e.*, a non-cancer endpoint) was the basis for the drinking water reference concentration, and both chronic and subchronic/short-term exposure values were available from the most recent resource, the lower concentration was used. Review the references and footnotes in [Table 2](#) for additional health effects information (*e.g.*, additional subchronic or chronic values).
- (5) For the contaminants that do not have a drinking water reference concentration available from a resource listed above, the RfDs from finalized health assessments are provided in [Table 2](#), if available. If a health assessment is not final, a link to additional information about its status is provided in a footnote.

Table 2. UCMR 5 Minimum Reporting Levels (MRLs), Regulatory Levels, and Health-Based Reference Values as of July 2026

Contaminant	[Note: To convert to ng/L or parts per trillion (ppt), multiply by 1,000]		Non-Regulatory Health-Based Reference Values		References
	UCMR 5 MRL (µg/L) ¹	Regulatory Level (µg/L)	Reference Concentration (µg/L)	RfD (mg/kg-day)	
lithium	9	-	HRL ² = 10	subchronic and chronic provisional RfD = 2×10^{-3}	Technical Support Document for the Final CCL 5 - Contaminant Information Sheets
PFOS	0.004	MCL ³ = 0.0040	-	-	2024 Final PFAS NPDWR
PFOA	0.004	MCL ³ = 0.0040	-	-	2024 Final PFAS NPDWR
HFPO-DA (GenX chemicals) ⁴	0.005	MCL ³ = 0.01	-	-	2024 Final PFAS NPDWR
PFHxS ⁴	0.003	MCL ³ = 0.01	-	-	2024 Final PFAS NPDWR
PFNA ⁴	0.004	MCL ³ = 0.01	-	-	2024 Final PFAS NPDWR
PFBS ^{4,5}	0.003	-	-	-	2024 Final PFAS NPDWR
PFBA	0.005	-	HBSL ⁶ = 6	chronic RfD = 1×10^{-3} subchronic RfD = 6×10^{-3}	2024 USGS Health-Based Screening Level
PFHxA	0.003	-	HBSL ⁶ = 3	subchronic and chronic RfD = 5×10^{-4}	2024 USGS Health-Based Screening Level
PFDA	0.003	-	-	subchronic and chronic RfD = 2×10^{-9}	2024 IRIS Assessment
6:2 FTS	0.005	-	-	chronic RfD = 2×10^{-4}	2024 Human Health Toxicity Value Assessment

¹ UCMR MRL – EPA-established UCMR Minimum Reporting Level. Based on laboratory capability; not related to contaminant health effects information.

² HRL – Health Reference Level calculated as part of the CCL 5 process. See [Terms and Definitions](#). The HRL for lithium is based on the RfD from the following health assessment: [Provisional Peer-Reviewed Toxicity Values for Lithium \(2008\)](#). For more information, see the EPA's [Technical Fact Sheet on Lithium in Drinking Water](#) for primacy agencies.

³ MCL – Maximum Contaminant Level. See [Terms and Definitions](#) and the PFAS [Final NPDWR](#) and [NPDWR Implementation](#) websites.

⁴ The agency proposed to rescind the April 2024 regulations for HFPO-DA (commonly known as GenX), PFHxS, PFNA, and the Hazard Index (HI) mixture of these three PFAS plus PFBS. See the [Proposed PFAS NPDWR Rescission Rule](#) website.

⁵ PFBS has a Health-Based Water Concentration (HBWC) of 2 µg/L associated with the April 2024 PFAS NPDWR. PFBS does not have an individual MCL; it is included in the Hazard Index (HI) calculation when PFBS occurs in a mixture with one or more of HFPO-DA, PFHxS, and PFNA. See [Terms and Definitions](#) and the PFAS [Final NPDWR](#) and [NPDWR Implementation](#) websites.

⁶ HBSL – USGS Health-Based Screening Level. See [Terms and Definitions](#). The HBSLs are based on the RfDs from the EPA's IRIS Assessments for [PFBA](#) and [PFHxA](#).

Contaminant	[Note: To convert to ng/L or parts per trillion (ppt), multiply by 1,000]		Non-Regulatory Health-Based Reference Values		References
	UCMR 5 MRL (µg/L) ¹	Regulatory Level (µg/L)	Reference Concentration (µg/L)	RfD (mg/kg-day)	
8:2 FTS	0.005	-	-	-	-
4:2 FTS	0.003	-	-	-	-
ADONA	0.003	-	-	-	-
11Cl-PF3OudS	0.005	-	-	-	-
9Cl-PF3ONS	0.002	-	-	-	-
NFDHA	0.02	-	-	-	-
PFEESA	0.003	-	-	-	-
PFMPA	0.004	-	-	-	-
PFMBA	0.003	-	-	-	-
PFDoA	0.003	-	-	-	-
PFHpS	0.003	-	-	-	-
PFHpA	0.003	-	-	-	-
PFPeS	0.004	-	-	-	-
PFPeA	0.003	-	-	-	-
PFUnA	0.002	-	-	-	-
NEtFOSAA	0.005	-	-	-	-
NMeFOSAA	0.006	-	-	-	-
PFTA	0.008	-	-	-	-
PFTrDA	0.007	-	-	-	-

Table 3. July 2026 UCMR 5 Data Summary for Unregulated Contaminants¹

Contaminant	UCMR 5 MRL ² (µg/L)	Ref Conc ³ (µg/L)	Total number of results	Number of results ≥MRL	Number of results >Ref Conc ⁴	% of results >Ref Conc	Total number of PWSs with results	Number of PWSs with results ≥MRL	Number of PWSs with results >Ref Conc ⁴	% of large PWSs (>10,000) with results >Ref Conc	% of small PWSs (≤10,000) with results >Ref Conc
lithium	9	10	66,675	18,939	13,134	19.7%	10,313	3,773	2,747	24.1%	28.5%
PFBA	0.005	6	66,356	5,463	0	0%	10,307	1,935	0	0%	0%
PFHxA	0.003	3	66,357	6,060	0	0%	10,307	1,807	0	0%	0%
PFDA	0.003	-	66,360	31	-	-	10,307	17	-	-	-
11CI-PF3OUdS	0.005	-	66,362	0	-	-	10,307	0	-	-	-
8:2 FTS	0.005	-	66,352	12	-	-	10,307	10	-	-	-
4:2 FTS	0.003	-	66,361	3	-	-	10,307	3	-	-	-
6:2 FTS	0.005	-	66,348	253	-	-	10,307	183	-	-	-
ADONA	0.003	-	66,361	4	-	-	10,307	3	-	-	-
9CI-PF3ONS	0.002	-	66,343	1	-	-	10,307	1	-	-	-
NFDHA	0.02	-	66,355	5	-	-	10,307	4	-	-	-
PFEESA	0.003	-	66,361	0	-	-	10,307	0	-	-	-
PFMPA	0.004	-	66,361	3	-	-	10,307	2	-	-	-
PFMBA	0.003	-	66,361	2	-	-	10,307	1	-	-	-
PFDoA	0.003	-	66,351	4	-	-	10,307	4	-	-	-
PFHpS	0.003	-	66,361	6	-	-	10,307	5	-	-	-
PFHpA	0.003	-	66,361	1,514	-	-	10,307	555	-	-	-
PFPeS	0.004	-	66,361	99	-	-	10,307	50	-	-	-
PFPeA	0.003	-	66,343	7,040	-	-	10,307	2,066	-	-	-
PFUnA	0.002	-	66,358	12	-	-	10,307	6	-	-	-
NEtFOSAA	0.005	-	66,603	1	-	-	10,309	1	-	-	-
NMeFOSAA	0.006	-	66,604	2	-	-	10,309	1	-	-	-
PFTA	0.008	-	66,604	0	-	-	10,309	0	-	-	-
PFTrDA	0.007	-	66,603	0	-	-	10,309	0	-	-	-

¹ This data summary represents all results that the EPA received for UCMR 5 monitoring. Analytical results from the UCMR program are reported by laboratories and provided by the agency in micrograms/liter (µg/L, or parts per billion). To convert results in µg/L to nanograms/liter (ng/L, or parts per trillion), multiply the value by 1,000. The UCMR results represented by this table are single measurements and do not represent a running annual average (RAA). For information on results for the NPDWR PFAS, see [Table 4](#). The total number of results and total number of PWSs with results for the NPDWR PFAS are similar to the other EPA Method 533 PFAS (*i.e.*, approximately 66,350 results from approximately 10,300 PWSs).

² UCMR MRL – EPA-established UCMR Minimum Reporting Level. Based on laboratory capability; not related to contaminant health effects information.

³ Ref Conc – Reference Concentration. Based on contaminant health effects information; non-regulatory and non-enforceable. The EPA CCL 5 Health Reference Level (HRL) for lithium and the USGS Health-Based Screening Levels (HBSLs) for PFBA and PFHxA. See [Terms and Definitions](#).

⁴ The HRL and the HBSLs are expressed with one significant digit; comparison of UCMR results to those levels is therefore based on one significant digit. Results >15 µg/L for lithium round to 20 µg/L; results >6.5 µg/L for PFBA round to 7 µg/L; and results >3.5 µg/L for PFHxA round to 4 µg/L and are identified as above the reference concentrations.

Comparing UCMR 5 Results to the April 2024 PFAS National Primary Drinking Water Regulation (NPDWR)

UCMR 5 results for the NPDWR PFAS do not indicate compliance or noncompliance with the MCLs.

Compliance with the PFAS MCLs will be determined by calculating the running annual average (RAA) of quarterly results for each sample point based on compliance monitoring data reported to the primacy agency. For more information on the MCLs and requirements, visit the PFAS [Final NPDWR](#) and [NPDWR Implementation](#) websites.

Recognizing that the UCMR 5 results have no direct implications for PWS MCL compliance, but that many are interested in comparing UCMR 5 data to the MCLs, the EPA calculated RAAs for the UCMR 5 results where such a calculation was possible. The agency then compared those averages to the MCLs. The UCMR 5 dataset represents approximately 10,130 PWSs with full sets of results for approximately 25,950 sampling locations (*i.e.*, two sample results for locations with ground water sources or four sample results for locations with surface water sources).

[Table 4](#) presents the UCMR 5 results using sampling location averages and the MCLs from the PFAS NPDWR. The approximate percentages of PWSs in this group who had one or more sampling location averages greater than the MCL for an individual PFAS or the Hazard Index (HI) are:

- 7.1% of the small PWSs,
- 8.3% of the medium PWSs, and
- 15.5% of the large PWSs.

Note: if a particular PWS had averages greater than multiple MCLs, that PWS was counted in this table multiple times (*i.e.*, once in each row for the respective MCLs). If this is not considered by those assessing the data in these rows, it would result in “double counting.” For more information, see the EPA’s [PFAS NPDWR Implementation website](#).

Based on the data summarized in [Table 4](#), a weighted estimate of the total PWSs nationwide with one or more averages greater than a PFAS NPDWR MCL is **7.8%**. This weighted estimate is provided since the UCMR 5 dataset includes only a subset of the PWSs serving fewer than 3,300 people that are subject to the requirements of the PFAS NPDWR. Therefore, a weighted estimate ensures the PWS size categories are appropriately represented relative to the amount of PWSs nationally and PWSs serving 3,300 or more people are not overrepresented.

Table 4. July 2026 Comparison of UCMR 5 Averages and the MCLs from the April 2024 NPDWR

The agency proposed to rescind the regulations for HFPO-DA (commonly known as GenX), PFHxS, PFNA, and the Hazard Index (HI) mixture of these three PFAS plus PFBS. Visit the EPA's [Proposed PFAS NPDWR Rescission Rule](#) website for more information.

PFAS in April 2024 NPDWR	MCL (µg/L) ¹	UCMR 5 PWSs with full set(s) of results ²			UCMR 5 PWSs with average(s) greater than MCL			% UCMR 5 PWSs with average(s) greater than MCL ³		
		large PWS (>10,000)	medium PWS ⁴ (3,300–10,000)	small PWS (<3,300)	large	medium	small	large	medium	small
PFOS	0.0040	4,275	5,077	775	523	307	45	12.2%	6.0%	5.8%
PFOA	0.0040	4,276	5,077	775	472	273	34	11.0%	5.4%	4.4%
HFPO-DA (GenX chemicals)	0.01	4,277	5,077	775	3	5	0	0.1%	0.1%	0.0%
PFHxS	0.01	4,273	5,077	775	40	22	4	0.9%	0.4%	0.5%
PFNA	0.01	4,276	5,077	775	3	6	0	0.1%	0.1%	0.0%
Hazard Index (HI) (HFPO-DA, PFHxS, PFNA, PFBS)	1 (unitless)	4,273	5,076	774	45	31	4	1.1%	0.6%	0.5%
% UCMR 5 large PWSs (serving >10,000) with average(s) greater than MCL = 15.5% (662/4,277)										
% UCMR 5 medium PWSs ⁴ (serving 3,300–10,000) with average(s) greater than MCL = 8.3% (419/5,077)										
% UCMR 5 small PWSs (serving <3,300) with average(s) greater than MCL = 7.1% (55/775)										
Estimated weighted % of PWSs nationwide with average(s) greater than MCL ⁵ = 7.8%										

¹ MCLs for PFOA and PFOS are expressed with two significant digits; MCLs for HFPO-DA, PFHxS, PFNA, and the HI are expressed with one significant digit. Comparison of UCMR 5 averages to MCLs is based on the corresponding number of significant digits. An average is counted as greater than the MCL for PFOS or PFOA if it is ≥0.00405 µg/L. An average is counted as greater than the MCL for HFPO-DA, PFHxS, or PFNA if it is ≥0.015 µg/L. An average is counted as greater than the HI MCL if it is ≥1.5 and at least two PFAS included in the average are measured at or above the UCMR 5 MRL.

² For UCMR 5, PWSs may have multiple sampling locations, for which there are either two or four sample events over a period of 12 months (for locations with ground water or surface water sources, respectively). PWSs were only considered in the comparison to each MCL if they have at least one corresponding full set of UCMR 5 results (*i.e.*, two or four results). Note: Compliance with the PFAS NPDWR MCLs will be determined by calculating the running annual average (RAA) of four quarterly results for each sample point. When calculating averages using UCMR 5 results for comparison with the MCLs, the average for locations with ground water sources is based on two semi-annual results and the average for locations with surface water sources is based on four quarterly results.

³ Percentages are based on data received from the PWSs subject to UCMR 5 monitoring. UCMR 5 included all CWSs and NTNCWSs serving 3,300 or more people and a nationally representative sample of CWSs and NTNCWSs serving fewer than 3,300 people. The PFAS NPDWR includes all CWSs and NTNCWSs.

⁴ The Safe Drinking Water Information System Federal Reporting Services (SDWIS/Fed) database and PFAS NPDWR Economic Analysis defines medium PWSs as those serving 3,301 to 10,000 people.

⁵ An estimated number of PWSs nationwide with average(s) >MCL in each PWS size category was calculated by multiplying the UCMR 5 percentage by the number of PWSs (*i.e.*, CWSs and NTNCWSs) in SDWIS/Fed. From that, an estimated weighted percentage of PWSs nationwide with average(s) >MCL was calculated by summing the estimated number of PWSs with averages(s) >MCL in each PWS size category, dividing by the total PWSs in SDWIS/Fed, and multiplying by 100.

UCMR 5 PFAS Co-occurrence in Drinking Water

The EPA's examination of drinking water data shows that different PFAS can often be found together and in varying combinations as mixtures (for additional discussion, see the EPA's [PFAS Occurrence and Contaminant Background Support Document](#) and the [final PFAS NPDWR](#)). Additionally, decades of research show mixtures of different chemicals can have additive health effects, even if the individual chemicals are each present at lower levels. The EPA established drinking water standards for certain PFAS to provide health protection against these individual and co-occurring PFAS in PWSs. In cases where the PFAS included in the final PFAS NPDWR occur at concentrations above their respective regulatory standards, there is also an increased probability of co-occurrence of additional unregulated PFAS.

[Table 5](#) provides a general assessment of UCMR 5 PFAS co-occurrence by location (*i.e.*, considering all available PFAS results for a sampling location). For UCMR 5 monitoring, PWSs may have had multiple sampling locations. Each sample event included sample collection for 29 PFAS. Results were included in the location-level counts regardless of whether results were available for all 29 PFAS for a sampling location. This table presents the count of sampling locations for which "N or more" unique PFAS were found at or above the UCMR 5 MRL, where N is 1-13 in Table 5. The table stops at 13 (and not 29) because 13 was the maximum number of unique PFAS found to co-occur at a sampling location (*i.e.*, same PWS, facility, and sample point).

Table 5. July 2026 UCMR 5 PFAS Co-occurrence Counts by Sampling Location

Number (N) of Unique PFAS ≥UCMR 5 MRL ¹	1	2	3	4	5	6	7	8	9	10	11	12	13
Number of Sampling Locations with N or More PFAS ²	6,499	4,286	3,183	2,321	1,614	1,092	676	338	90	21	6	1	1
Number of Unique PWSs Associated with Locations ³	3,595	2,416	1,874	1,431	1,006	696	432	224	69	18	6	1	1

¹ Represents the number (N) "or more" count of unique PFAS found at or above their UCMR 5 Minimum Reporting Level (MRL). UCMR MRLs are based on laboratory capability and are not related to contaminant health effects information.

² Represents the number of individual sampling locations with the corresponding number of unique PFAS found across all available results for the location. For example, 3,183 sampling locations each had three or more different PFAS occur; these 3,183 sampling locations were from 1,874 unique PWSs.

³ Represents the number of unique PWSs associated with the number of sampling locations for each count. For example, 1,874 different PWSs each had at least one sampling location with three or more different PFAS occurrences.

Terms and Definitions

- a) **HBSL** – USGS non-cancer Health-Based Screening Level. [HBSLs](#) are non-enforceable water quality benchmark concentrations of contaminants in water developed using the latest [EPA methods](#) for establishing drinking water guidelines and the most recent EPA peer-reviewed toxicity information. Non-cancer HBSLs are maximum concentrations that are not expected to cause adverse non-carcinogenic health effects over a lifetime of exposure.
- b) **HBWC** – Health-Based Water Concentration. The level below which there are no known or anticipated adverse health effects over a lifetime of exposure, including for sensitive populations and life stages, and allows for an adequate margin of safety.
- c) **HI** – Hazard Index. A long-established approach that the EPA regularly uses to understand health risk from a chemical mixture (*i.e.*, exposure to multiple chemicals). The HI is made up of a sum of fractions. Each fraction compares the level of each PFAS measured in the water to the HBWC. The HI is the sum of component hazard quotients, which are calculated by dividing the measured PFAS component contaminant concentration in water [*e.g.*, expressed as parts per trillion (ppt) or nanograms per liter (ng/L)] by the associated HBWC expressed in the same units as the measured concentration (*e.g.*, ppt or ng/L). The HI MCL for the April 2024 PFAS NPDR is set at 1 (unitless).
- d) **HRL** – Health Reference Level. Derived during the EPA’s Contaminant Candidate List (CCL) process for screening purposes. HRLs are used in the EPA’s Regulatory Determination process as risk-derived concentrations against which to evaluate occurrence data to determine if contaminants occur at levels of public health concern. HRLs are not final determinations about the level of a contaminant in drinking water that is necessary to protect any particular population and, in some cases, are derived prior to development of a complete exposure assessment using the best available data. HRLs are not legally enforceable federal standards. To determine the HRL for a chemical, the agency considers adverse health effects that may pose a greater risk to specific life stages and other sensitive groups which represent a meaningful portion of the population. For more information on HRL derivation, see the [Technical Support Document for the Final CCL 5 – Contaminant Information Sheets](#).
- e) **MCL** – Maximum Contaminant Level. The highest level of a contaminant that is allowed in drinking water. MCLs are set as close to MCLGs as feasible using the best available treatment technology and taking cost into consideration. MCLs are enforceable standards.
- f) **MCLG** – Maximum Contaminant Level Goal. The level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety and are non-enforceable public health goals.
- g) **Ref Conc** – Health-Based Drinking Water Reference Concentration. Based on publicly available health information found in the following resources: EPA CCL 5 Contaminant Information Sheets [*i.e.*, Health Reference Levels (HRLs)], USGS Health-Based Screening Levels (HBSLs). Reference concentrations are derived from peer-reviewed health assessments published by the EPA or other governmental agencies. They are not legally enforceable federal standards and are subject to change as new health assessments are completed. Depending on available health effects information, a reference concentration in drinking water can be derived from a reference dose (RfD) (*i.e.*, a non-cancer endpoint) or a cancer slope factor (CSF) (*i.e.*, a cancer endpoint), and considers additional assumptions about body weight and drinking water intake.
- h) **RfD** – Oral Reference Dose. A non-cancer estimate (with uncertainty spanning perhaps an order of magnitude) of a daily oral exposure to the human population (including sensitive subgroups) that is likely to be without an appreciable risk of deleterious effects during a lifetime. It is typically derived by dividing a point-of-departure (POD) from a selected dose-response study (*e.g.*, no-observed-adverse-effect level [NOAEL], lowest-observed-adverse-effect level [LOAEL], benchmark dose [BMD]) by the uncertainty factors (UFs) applied to reflect limitations of the data used. Chronic RfDs are typically derived from animal toxicological studies with an exposure duration of months to years, representing a lifetime exposure in humans. Subchronic RfDs are typically derived from animal toxicological studies with an exposure duration of 31 to 90 days, representing a less than lifetime exposure in humans (up to 10% of average lifespan). Visit the [EPA’s IRIS website](#) for more information about RfD derivation.
- i) **UCMR MRL** – EPA-established UCMR Minimum Reporting Level. The lowest concentration that laboratories may report to the EPA during UCMR monitoring. MRLs are not associated with health effects information. More specifically, an MRL is the quantitation limit for a contaminant that is considered achievable, with 95% confidence, by at least 75% of laboratories nationwide using a specified analytical method (recognizing that individual laboratories may be able to measure at lower levels). **[Note: The Agency for Toxic Substances and Disease Registry (ATSDR) uses the term “MRL” for a different purpose (*i.e.*, to describe “minimal risk level”). The UCMR term and the ATSDR term have no relationship to each other.]**

Data Considerations

The UCMR 5 analytical results are publicly available through the [UCMR Occurrence Data](#) website.

The UCMR Data Finder allows people to easily search for, summarize, and download the UCMR analytical results. Results can be filtered using multiple data fields, including PWS, PWS size, state, EPA Region, contaminant, source water type, and results at or above UCMR MRLs (data definitions provided in [Table 6](#)). The UCMR Data Finder is available to federal, state, and local agencies as well as the public to easily locate and retrieve specific results and assist with answering questions regarding UCMR monitoring. Selected results can be viewed online or downloaded as a Microsoft Excel file (.xlsx). A video demonstration of the UCMR Data Finder is available.

For those interested in large-scale data processing using statistical or data analysis software, the EPA recommends using the occurrence data text files containing the UCMR 5 analytical results as well as additional information reported during monitoring. Data are provided in tab delimited text files (.txt) (see below for descriptions), with field names included in the first row of each file and no text qualifier. The EPA recommends importing all ID fields into your choice of software as text since some of the IDs can otherwise be misinterpreted as long integer field types when they contain alpha characters.

- To download the occurrence data text files (data definitions provided in [Table 7](#)), select one of the following compressed (zipped) folders (.zip) from [UCMR 5 \(2023-2025\) Occurrence Data](#):
 - **UCMR 5 Occurrence Data Text Files** (.zip) to view all the analytical results (*i.e.*, results for all contaminants reported by all PWSs). The **UCMR5_All.txt** file may be too large to be imported into Excel, in which case you can try other applications (*e.g.*, Microsoft Access) or import a subset of the data as described in the following two bullets.
 - **UCMR 5 Occurrence Data Text Files by State** (.zip) to view all the analytical results, organized by Tribes and states. Within this zip folder, one text file (**UCMR5_All_Tribes_AK_LA.txt**) will have all results for Tribal PWSs and for the states starting alphabetically with A through L; another file (**UCMR5_All_MA_WY.txt**) will have all results for the states starting alphabetically with M through W. The results are organized this way to address file size limitations and streamline data management.
 - **UCMR 5 Occurrence Data Text Files by Method Classification** (.zip) to view all the analytical results, organized by analytical method. Within this zip folder, you will find individual text files with results organized by method (*e.g.*, a Method 200.7 file with results for lithium).
- Each of the compressed (zipped) folders above also contain the following text files for **additional data elements** (*i.e.*, information beyond analytical results for the 30 UCMR 5 contaminants):
 - **UCMR5_ZIPCodes.txt** – U.S. Postal Service ZIP Code(s) for all areas served by a PWS (data definitions provided in [Table 8](#))
 - **UCMR5_AddtlDataElem.txt** – Disinfectant Type, Treatment Information, Lithium Occurrence, Lithium Treatment, PFAS Occurrence, PFAS Treatment, Potential PFAS Sources, Potential PFAS Sources Detail (data definitions provided in [Table 9](#))

For step-by-step details, refer to the document [Instructions for Accessing UCMR Results](#). Additional reference material is available on the [UCMR 5 website](#).

Table 6. Data Definitions for the UCMR Data Finder

Field Name	Definition
PWS ID	Public Water System (PWS) Identification Code. The code used to identify each PWS. The code begins with the standard 2-character postal state abbreviation or Region code for Tribes; the remaining 7 numbers are unique to each PWS in the state. Utah PWS IDs begin with 4 letters (UTAH) followed by 5 numbers
PWS Name	Name of the PWS
Contaminant	The UCMR 5 contaminant analyzed
Result (µg/L)	Numeric value of the analytical result in µg/L for the contaminant. Results less than the UCMR 5 MRL are indicated by <MRL
Collection Date	Date of sample collection (month, day, year)
Facility ID	Identification code for each applicable facility associated with water treatment or delivery at the PWS
Facility Name	Name of the facility at the PWS
Sample Point ID	Identification code for each sample point location at the PWS
Sample Point Name	Name of the sample point at the PWS
Sample Event Code	Identification code for each sample event: SE1, SE2, SE3, SE4
Sample ID	Identification code for each sample
Method ID	Identification code of the analytical method
PWS Size	Size category of the PWS for UCMR 5, based on retail population as indicated by the Safe Drinking Water Information System (Federal) (SDWIS/FED) as of February 1, 2021: S (≤ 10,000), L (> 10,000)
Facility Water Type	Source of water at the facility: SW (surface water), GW (ground water), GU (ground water under the direct influence of surface water), MX (any combination of SW, GW, and GU)
Sample Point Type	Sampling Point Type Code: EP (entry point to the distribution system)
EPA Region	EPA Region (states): Region 1 (CT, ME, MA, NH, RI, VT), Region 2 (NJ, NY, PR [Puerto Rico], VI [Virgin Islands]), Region 3 (DE, DC, MD, PA, VA, WV), Region 4 (AL, FL, GA, KY, MS, NC, SC, TN), Region 5 (IL, IN, MI, MN, OH, WI), Region 6 (AR, LA, NM, OK, TX), Region 7 (IA, KS, MO, NE), Region 8 (CO, MT, ND, SD, UT, WY), Region 9 (AZ, CA, HI, NV, AS [American Samoa], GU [Guam], MP [Northern Marianas Islands], NN [Navajo Nation]), Region 10 (AK, ID, OR, WA)
State	State abbreviation. Tribal PWSs without primacy are attributed to an EPA Region (01, 02, 03, 04, 05, 06, 07, 08, 09, 10)
UCMR Minimum Reporting Level (MRL, µg/L)	Minimum Reporting Level defined by UCMR 5 in µg/L for the contaminant. Based on laboratory capability; not related to contaminant health effects information (see Terms and Definitions)

Table 7. Data Definitions for Text Files: UCMR5_All, UCMR5_All_Tribes_AK_LA, UCMR5_All_MA_WY, and UCMR5_MethodNumber

Field Name	Definition
PWSID	Public Water System (PWS) Identification Code. The code used to identify each PWS. The code begins with the standard 2-character postal state abbreviation or Region code for Tribes; the remaining 7 numbers are unique to each PWS in the state. Utah PWS IDs begin with 4 letters (UTAH) followed by 5 numbers
PWSName	Name of the PWS
Size	Size category of the PWS for UCMR 5, based on retail population as indicated by the Safe Drinking Water Information System (Federal) (SDWIS/FED) as of February 1, 2021: S ($\leq 10,000$), L ($> 10,000$)
FacilityID	Identification code for each applicable facility associated with water treatment or delivery at the PWS
FacilityName	Name of the facility at the PWS
FacilityWaterType	Source of water at the facility: SW (surface water), GW (ground water), GU (ground water under the direct influence of surface water), MX (any combination of SW, GW, and GU)
SamplePointID	Identification code for each sample point location at the PWS
SamplePointName	Name of the sample point at the PWS
SamplePointType	Sampling Point Type Code: EP (entry point to the distribution system)
AssociatedFacilityID	Null for UCMR 5
AssociatedSamplePointID	Null for UCMR 5
CollectionDate	Date of sample collection (month, day, year)
SampleID	Identification code for each sample
Contaminant	The UCMR 5 contaminant analyzed
MRL	Minimum Reporting Level (MRL) defined by UCMR 5 in $\mu\text{g/L}$ for the contaminant. Based on laboratory capability; not related to contaminant health effects information (see Terms and Definitions)
Units	Units of the UCMR 5 MRL and analytical results: $\mu\text{g/L}$
MethodID	Identification code of the analytical method
AnalyticalResultsSign	Sign indicating whether the analytical result is less than (<) the UCMR 5 MRL or equal to (=) a numeric value at or above the UCMR 5 MRL
AnalyticalResultValue	Numeric value of the analytical result in $\mu\text{g/L}$ for the contaminants. Null (or blank) values represent results less than the UCMR 5 MRL
SampleEventCode	Identification code for each sample event: SE1, SE2, SE3, SE4
MonitoringRequirement	AM (Assessment Monitoring)
Region	EPA Region (states): 1 (CT, ME, MA, NH, RI, VT), 2 (NJ, NY, PR [Puerto Rico], VI [Virgin Islands]), 3 (DE, DC, MD, PA, VA, WV), 4 (AL, FL, GA, KY, MS, NC, SC, TN), 5 (IL, IN, MI, MN, OH, WI), 6 (AR, LA, NM, OK, TX), 7 (IA, KS, MO, NE), 8 (CO, MT, ND, SD, UT, WY), 9 (AZ, CA, HI, NV, AS [American Samoa], GU [Guam], MP [Northern Marianas Islands], NN [Navajo Nation]), 10 (AK, ID, OR, WA)
State	State abbreviation. Tribal PWSs without primacy are attributed to an EPA Region (01, 02, 03, 04, 05, 06, 07, 08, 09, 10)
UCMR1SampleType	Null for UCMR 5

Table 8. Data Definitions for Text File: UCMR5_ZIPCodes

Field Name	Definition
ZIPCODE	U.S. Postal Service ZIP Code(s) for all areas served by a PWS. This is entered by the PWS

Table 9. Data Definitions for Text File: UCMR5_AddtlDataElem

Additional Data Element	Definition and Response Options
DisinfectantType	All of the disinfectants/oxidants that have been added prior to and at the entry point to the distribution system. Please select ALL that apply. PEMB = permanganate, HPXB = hydrogen peroxide, CLGA = gaseous chlorine, CLOF = offsite generated hypochlorite (stored as a liquid form), CLON = onsite generated hypochlorite, CAGC = chloramine (formed with gaseous chlorine), CAOF = chloramine (formed with offsite hypochlorite), CAON = chloramine (formed with onsite hypochlorite), CLDB = chlorine dioxide, OZON = ozone, ULVL = ultraviolet light, OTHD = other types of disinfectant/oxidant, NODU = no disinfectant/oxidant used
TreatmentInformation	Treatment information associated with the sample point. Please select ALL that apply. CON = conventional (non-softening, consisting of at least coagulation/sedimentation basins and filtration), SFN = softening, RBF = river bank filtration, PSD = pre-sedimentation, INF = in-line filtration, DFL = direct filtration, SSF = slow sand filtration, BIO = biological filtration (operated with an intention of maintaining biological activity within filter), UTR = unfiltered treatment for surface water source, GWD = ground water system with disinfection only, PAC = application of powder activated carbon, GAC = granular activated carbon adsorption (not part of filters in CON, SFN, INF, DFL, or SSF), AIR = air stripping (packed towers, diffused gas contactors), POB = pre-oxidation with chlorine (applied before coagulation for CON or SFN plants or before filtration for other filtration plants), MFL = membrane filtration, IEX = ionic exchange, DAF = dissolved air floatation, CWL = clear well/finished water storage without aeration, CWA = clear well/finished water storage with aeration, ADS = aeration in distribution system (localized treatment), OTH = other types of treatment, NTU = no treatment used, DKN = do not know
LithiumOccurrence	A yes or no answer provided by the PWS for each entry point to the distribution system. Question: Have you (or your wholesaler) tested for lithium in your drinking water in the past, prior to monitoring for UCMR 5? YES = If yes, did you modify your treatment and if so, what types of treatment did you implement? (see LithiumTreatment); NO = have never tested for the contaminant; DK = do not know
LithiumTreatment	If yes, select ALL that apply: PAC = application of powder activated carbon, GAC = granular activated carbon adsorption (not part of filters in CON, SFN, INF, DFL, or SSF), IEX = ionic exchange, NRO = nanofiltration and reverse osmosis, OZN = ozone, BAC = biologically active carbon, MFL = membrane filtration, UVL = ultraviolet light, OTH = other, NMT = not modified after testing
PFASOccurrence	A yes or no answer provided by the PWS for each entry point to the distribution system. Question: Have you (or your wholesaler) tested for PFAS in your drinking water in the past, prior to monitoring for UCMR 5? YES = If yes, did you modify your treatment and if so, what types of treatment did you implement? (see PFASTreatment); NO = have never tested for the contaminant; DK = do not know
PFASTreatment	If yes, select ALL that apply: PAC = application of powder activated carbon, GAC = granular activated carbon adsorption (not part of filters in CON, SFN, INF, DFL, or SSF), IEX = ionic exchange, NRO = nanofiltration and reverse osmosis, OZN = ozone, BAC = biologically active carbon, MFL = membrane filtration, UVL = ultraviolet light, OTH = other, NMT = not modified after testing
PotentialPFASSources	A yes or no answer provided by the PWS for each entry point to the distribution system. Question: Are you aware of any potential current and/or historical sources of PFAS that may have impacted the drinking water sources at your water system? YES = If yes, select ALL that apply (see PotentialPFASSourcesDetail); NO = not aware of any potential current and/or historical sources; DK = do not know
PotentialPFASSourcesDetail	If yes, select ALL that apply: MB = military base, FT = firefighting training school, AO = airport operations, CW = car wash or industrial launderers, PS = public safety activities (e.g., fire and rescue services), WM = waste management, HW = hazardous waste collection, treatment, and disposal, UW = underground injection well, SC = solid waste collection, combustors, incinerators, MF = manufacturing, FP = food packaging, TA = textile and apparel (e.g., stain- and water-resistant, fiber/thread, carpet, house furnishings, leather), PP = paper, CC = chemical, PR = plastics and rubber products, MM = machinery, CE = computer and electronic products, FM = fabricated metal products (e.g., nonstick cookware), PC = petroleum and coal products, FF = furniture, OG = oil and gas production, UT = utilities (e.g., sewage treatment facilities), CT = construction (e.g., wood floor finishing, electrostatic painting), OT = other